Science & Technology

70RSAT26RFI000017
ACE-CASPER Experiment 2026

Advanced Communications Experiment – Cross-border Autonomous Vehicle Session Persistence Experiment and Research

Submission Deadline: May 8, 2026 – 5:00 p.m. EST

Table of Contents

Quick Start Guide

✓ Who can apply: Organizations with TRL 6+ technologies in UAS or AV

✓ What to submit: Completed form, plus optional whitepaper (5 pages).

✓ Where to send: NGFR@hq.dhs.gov, subject line: “(Org Name) Application – 2026 ACE-CASPER Experiment”.

✓ Deadline: **May 8, 2026, 5:00 p.m. EST.**

✓ Participation offers respondents a unique opportunity to showcase technology to DHS S&T, Canada’s DRDC, and state and local public safety stakeholders across North America.

✓ Key Dates: Prototype due **May 29, 2026;** Dry Run **Late Fall 2026**; Final Experiment **November 2026.**

Background

The U.S. Department of Homeland Security (DHS) Science and Technology Directorate's (S&T), in collaboration with Defence Research and Development Canada, Centre for Security Science (DRDC CSS), is hosting the 2026 Advanced Communications Experiment Cross-border Autonomous Vehicle Session Persistence Experiment and Research (ACE-CASPER). This multi-day experiment will take place along the U.S.-Canada border and will simulate a national emergency response scenario, evaluating how autonomous vehicles (AV), unmanned aerial systems (UAS), and 5G-enabled technologies can support persistent cross-border communications, enhance situational awareness, and improve coordination between U.S. and Canadian response teams. During the experiment, UAS and AV will move across both sides of the border streaming live video to a bi-national command and control center (C2), deploy sensor-based payloads to assess the situation, relay that data back to C2, and use that data to inform decision making and response. The experiment will demonstrate:

- Secure, resilient 5G session persistence, with connectivity across U.S. and Canadian border networks;
- Seamless command and control (C2) handoff, and real-time data interoperability;
- Operational readiness validation for DHS components and DRDC with State, Provincial, and Local players in cross-border coordination; and
- Bi-national response coordination using enhanced, real-time situational awareness tools.

Vehicle autonomy level is secondary to the experiment's primary objective of demonstrating resilient, persistent 5G communications across U.S. and Canadian networks.

Objective

The ACE-CASPER Experiment aims to evaluate the operational readiness and technological capabilities of 5G Session Persistence and 5G connectivity with UAS and AV systems in a cross-border emergency scenario. Vendors are expected to demonstrate solutions that ensure secure, resilient 5G connectivity, real-time situational awareness, and seamless interoperability between U.S. and Canadian networks.

The experiment does not require autonomous decision-making or AI-driven navigation; the primary focus is validating reliable 5G connectivity, session persistence, and command-and-control performance.

Why Your Participation Matters

By joining this experiment, organizations can demonstrate technology capabilities in a high-profile, DHS S&T/DRDC-sponsored environment that mirrors real-world emergency and operational conditions. This unprecedented visibility and access to end users in an operational environment can strengthen the credibility and reliability of your solution with public safety agencies at the Federal, State, Local, and international levels. Participation in this experiment is not a guarantee of a future contract or procurement.

Key Benefits for Participating in the ACE-CASPER Bi-National Experiment:

- Industry influence
- Build key relationships
- Knowledge transfer
- Technology validation
- Identify interoperability considerations
- Identify research gaps or challenges for integration/operationalizing solutions

UAS Requirements

There are two categories of technology for which DHS S&T is seeking information. They are:

1) Unmanned Aerial Systems (UAS)

2) Autonomous Vehicles (AV)

Unmanned Aerial System (UAS)

DHS S&T is seeking to evaluate technology solutions that provide UAS operations for the ACE-CASPER Experiment. The UASs will be directly connected to the 5G networks as they move within and between both countries. UAS platforms may support single-SIM or multi-SIM configurations to enable connectivity across U.S. and Canadian 5G networks, depending on the vendor solution.

The UASs in the experiment will provide a live high-definition video feed accessible to all experiment participants in real time, UAS pilot operators must have FAA part 107 and Part 135 Certification to meet Federal Aviation Administration (FAA) compliance for the experiment.

For this experiment, the UAS must connect to the 5G network and support vertical takeoff. and landing (VTOL) to enable operations in constrained or infrastructure-limited environments typical of border and emergency response scenarios. There are three key components of the UAS system that include avionics, command and control, and detect- and-avoid (DAA) alerting.

The following list features capabilities which DHS S&T is seeking solutions:

Basic UAS Operational Capabilities:

1. 5G modem for network connectivity, command and control (C2), alerting, and use of sensor payloads
2. Sensor payloads may include, but are not limited to, electro-optical (EO) and infrared (IR) cameras, environmental sensors, or other mission-relevant payloads capable of supporting situational awareness and data sharing during the experiment.
3. Pilot operators must hold Federal Aviation Administration (FAA) Part 107 certification, and Part 135 certification where applicable, to meet FAA compliance requirements for the experiment.
4. Vertical take-off and landing
5. Up to 5 km (3 mile) range
6. Flight ceiling of 100 meters (330 feet)
7. Beyond visual line of sight (BVLOS) control
8. Payload delivery not to exceed 5lbs

<u>UAS Avionics Requirements</u>: Key components include but not limited:

1. 5G Modem for network connectivity (3GPP Release 18+ Stand Alone Modem)
    a. Common SA bands include sub-6 GHz low-band (n71/600MHz, n28/700MHz) for coverage, mid-band (n41/2.5GHz, n78/3.5GHz) for speed/capacity, and high-band/mmWave (n258/26GHz, n260/39GHz)
    b. Ability to swap modems if the modem fails to operate, or for failover
2. Airframe: UAV platform and structure
3. Propulsion: Motors, propellers, and power systems
4. Avionics: Electronic systems, sensors, and navigation
5. Communication: Data transmission and reception
6. Payload: Cameras, sensors, or cargo
7. UAS Battery Performance (UAS flight duration) minimum 30 minutes of flight time is preferred, including cold weather environments, typical of the U.S. Canadian border.

<u>UAS Command and Control (C2) Capabilities</u>: Encompasses the hardware, software, etc., that enable communications and remote management between UAS operators and the UAS, allowing for control of navigation, surveillance, payloads, etc., and overall experiment

execution. UAS will be provisioned to supply all of the following, but not limited to, key mission capabilities and features:

1. Command and Control: Intelligence Surveillance Reconnaissance (C2ISR)
2. Situational Awareness: Real-time surveillance for decision-making; relaying threats information; sharing critical information with low latency
3. Surveillance: Real-time jurisdictional intelligence
4. Reconnaissance: Tactical mission planning
5. Search and Rescue: Quick response, use of deployable assets
6. Aerial Videography/Photography: 4K High-quality imagery + Infrared imagery;
7. Disaster Response: Damage assessment, search and rescue
8. **Photography:** Capturing imagery to support search and rescue operations and terrain awareness, enabling operator-assisted assessment of the operational environment.

UAS Detect and Avoid (DAA) System Capabilities: provide timely warnings to the UAS operators about potential hazards, allowing them to take corrective action and avoid collisions. DAA is key to the experiment, being able to provide alerting levels, configurable alerts, remote ID, etc. Key components include but not limited:

1. Alerting System: alerts of other UAVs/Aircrafts within the coverage area
2. Alerting Levels and Location
3. Collision Avoidance Alerts: Proximity warnings for nearby aircraft or obstacles

AV Requirements

DHS S&T is seeking to evaluate technology solutions that provide autonomous vehicle (AV) operations supported by 5G connectivity. For the purposes of this experiment, “autonomous vehicle” refers to vehicles capable of autonomous, semi-autonomous, or remotely supervised operation, including beyond visual line-of-sight (BVLOS) control. For this experiment, AVs are expected to use 5G to enable command and control and support safe operations, with human operators maintaining oversight. The AVs will be connected to 5G networks as they move within and between both countries, allowing for session persistence, route coordination, alerting, and real-time data sharing. AVs will provide live high-definition video feeds accessible to all experiment participants, and vehicle operators will be responsible for ensuring safe operations during the experiment.

It is expected that the AV to provide terrestrial coverage of up to 5 kilometers (3 miles). The AVs will need to be controlled beyond visual line-of-sight (BVLOS). AVs will provide situational awareness and are able to deliver a small rescue package to an individual in distress that cannot be rescued immediately. The AVs will be able to deliver real-time video feeds to all participants in the experiment.

There are three key components of the AV system that are important to the experiment: autonomous vehicle avionics, command and control, and alerting.

<u>Autonomous Vehicle Avionics (AVA)</u>: Focus on the AV propulsion, vehicle management system, sensors, etc. Key AVA components should include, but not limited to:

1. 5G Modem for network connectivity (3GPP Release 18+ Stand Alone Modem)
2. AV Frame: AV platform and structure
3. Propulsion: Motors and power systems
4. AVA: Electronic systems, sensors, and navigation
5. Communication: Data transmission and reception
6. Payload (in weight) including but not limited to: Cameras, sensors, or deployable'
7. AV Battery life: Parameters around
8. AV battery life - Battery life as it relates to distance
9. Battery life as it relates to time during the scenarios/experiment
    9.1. Battery life: minutes/hours before battery has to be changed
10. Changing the battery during the experiment (including hot-swap or other resilience mechanisms)
11. Battery life relates to failover during experiment

<u>AV Command and Control (C2)</u>: Supports remote operation, supervision, and/or autonomous functions of AV navigation, sensing, and payload operations during the experiment. Key AV C2 components should include but not limited to:

1. Command and Control: Intelligence Surveillance Reconnaissance (C2ISR)
2. Situational Awareness: Real-time surveillance for decision-making
3. Surveillance: Real-time battlefield intelligence
4. Reconnaissance: Tactical mission planning

5. Search and Rescue: Quick response, use of deployable assets
6. Terrestrial Videography: 4K High-quality imagery + Infrared Imagery
7. Disaster Response: Damage assessment, search and rescue
8. Photography: Capturing search and rescue, local terrain, identifying potential threats, etc.

AV Detect and Avoid (DAA) Alerting System: Focuses on being able to provide timely warnings to the AV operators about potential hazards, allowing them to take corrective action and avoid collisions. Key DAA components should include but not limited to:

1. Alerting System: alerts of other AVs within the coverage area
2. Alerting Levels and Location/Configurable alerts
3. Collision Avoidance Alerts: Proximity warnings for nearby AVs, obstacles, etc.

General Participation Requirements

Prototypes of the technology solution hardware and/or software must be available **no later than May 29, 2026.**

Selected organizations will be required to enter into a Limited Purpose Cooperative Research and Development Agreement (LP-CRADA) with the Government.

- **This is NOT a contract award, and no money will be exchanged.**
- The purpose of the agreement is to define the conditions of participation (i.e., DHS S&T provides the test environment; the organization uses it and shares data in return. Intellectual property rights are agreed upon in advance, as security parameters for sharing information.
- DHS will not be liable for damage to technology solutions provided by respondents caused by their use in this experimentation, or any damage resulting from their interface or interaction with other organization's technology solutions or public safety communications systems provided by entities outside the Federal Government.

By participating, organizations gain the opportunity to:

- Demonstrate their technology in a real-world, bi-national operational environment with U.S. and Canadian agencies.
- Showcase capabilities directly to DHS S&T, DRDC, and frontline public safety users who influence future adoption and procurement decisions.

- Collect performance data and user feedback that can strengthen future product development, commercialization strategies, and requirements.
- Become a trusted partner in advancing resilient, interoperable communications for public safety and cross-border security.

This operational experiment is an integration event. All organizations that apply must understand that their technologies will be integrated into a larger system of first responder and border technology and will share data between devices owned and operated by other companies and/or first responder organizations. Hardware and/or software integration is required, and participating organizations will assist with that integration.

- Do not submit an application if you are unwilling to integrate your solution with other technologies.

- Collaboration on integration **will begin in Spring 2026** and will include an integration test event in **Late Fall 2026** in a laboratory environment to ensure that the integrated multi-vendor system works before it is shared with first responders during the dry run.

If participating organizations choose to do so, there will be an option to leave technology prototypes/licenses with the partnering first responder organizations for an extended evaluation period after the Experiment, of up to six months. DHS S&T will assist with coordinating additional testing and evaluation, and that data will be shared with the organization.

Your submission is not just an application, it's a chance to demonstrate your solution in a real-world, operational environment that shapes the future of public safety technology.

Participation will be limited and the priority will be given to:

- Organizations that fully complete this application form and thoroughly describe how their technology solution works and how it can integrate using open standards.

- Technology solutions that work with existing public safety infrastructure and communications systems. Solutions that require large, expensive communications systems to be replaced in order to achieve the desired effect are not practical for law enforcement and public safety use.

Submission Instructions

Submit completed forms (Attachment 1) to **NGFR@hq.dhs.gov** by **May 8, 2026, 5:00 p.m. EST.** Attach optional 5-page whitepaper. Mark proprietary information clearly.